Day 46 — Technical Question Hierarchy Model

1. Objective

The objective is to organize technical interview questions according to job role, relevant skills, experience level, and question difficulty.

2. Question Classification

Questions are organized into three experience categories.

Experience Category	Experience Range	Configured Difficulty
Entry	0–2 years	Basic
Intermediate	3–4.99 years	Intermediate
Advanced	5+ years	Advanced

3. Role-wise Skill Mapping

3.1 MERN Developer

1. MongoDB
1. Express.js
1. React
1. Node.js
1. JavaScript

3.2 Java Developer

1. Java
1. Object-Oriented Programming
1. Spring Boot
1. SQL
1. REST APIs

3.3 DevOps Engineer

1. Linux

1. Docker
1. Kubernetes
1. CI/CD
1. AWS
1. Monitoring

4. Question Categories

Category	Purpose
Experience-based	Understand previous technical work and practical experience
Conceptual	Evaluate understanding of technical concepts
Scenario-based	Evaluate approaches to practical technical situations

5. Sample Question Hierarchy

MERN Developer

Level	Skill	Difficulty	Sample Question
Entry	JavaScript	Basic	Explain the difference between var, let and const.
Intermediate	Node.js	Intermediate	Describe how you built and structured a REST API using Node.js.
Advanced	Node.js	Advanced	Explain how Node.js handles concurrency and asynchronous workloads.

Java Developer

Level	Skill	Difficulty	Sample Question
Entry	Java	Basic	Explain the main features of Java.
Intermediate	Spring Boot	Intermediate	Explain dependency injection in Spring Boot.
Advanced	Java	Advanced	Explain Java concurrency and thread synchronization.

DevOps Engineer

Level	Skill	Difficulty	Sample Question
Entry	Docker	Basic	What is a Docker image and how is it different from a container?
Intermediate	Kubernetes	Intermediate	Explain the role of Pods and Deployments in Kubernetes.
Advanced	Kubernetes	Advanced	Design a highly available Kubernetes deployment.

6. Question Selection Logic

The Technical Interview Engine:

1. Receives the technical role and experience in years.
2. Validates the role and experience input.
3. Determines the experience classification.
4. Loads the corresponding role-specific question set.
5. Returns the selected questions with skill and difficulty information.

7. Limitations

The current configuration provides initial sample questions for three roles.

The system selects questions using experience classification. It does not yet adapt the next question's difficulty dynamically according to the candidate's previous answer.

The question bank can be expanded through configuration updates.